

Terraform Complete Learning Scenario: Variables + terraform.tfvars (Car Movement Model)

This guide explains **different Terraform variable types** using one simple scenario: A car can move:

- Left
 - Right
 - Up (Forward)
 - Down (Reverse)
-

Part 1: `variables.tf`

Purpose:

Defines variable structure, type, defaults, and descriptions.

```
# =====  
# STRING VARIABLES  
# Each stores one movement direction  
# =====  
  
variable "l" {  
  type      = string  
  default   = "left-side"  
  description = "Car moves left"  
}  
  
variable "r" {  
  type      = string  
  default   = "right-side"  
  description = "Car moves right"  
}  
  
variable "u" {  
  type      = string  
  default   = "up-side"  
  description = "Car moves forward"  
}  
  
variable "d" {  
  type      = string  
  default   = "down-side"  
  description = "Car moves backward"
```

```

}

# =====
# NUMBER VARIABLES
# Controls movement speed
# =====

variable "left_speed" {
  type      = number
  default   = 20
  description = "Speed when moving left"
}

variable "right_speed" {
  type      = number
  default   = 20
  description = "Speed when moving right"
}

variable "up_speed" {
  type      = number
  default   = 60
  description = "Speed when moving forward"
}

variable "down_speed" {
  type      = number
  default   = 10
  description = "Speed when moving backward"
}

# =====
# BOOLEAN VARIABLES
# Enable or disable movement
# =====

variable "left_enable" {
  type      = bool
  default   = true
  description = "Enable left movement"
}

variable "right_enable" {
  type      = bool
  default   = true
  description = "Enable right movement"
}

variable "up_enable" {

```

```

    type      = bool
    default   = true
    description = "Enable forward movement"
}

variable "down_enable" {
    type      = bool
    default   = true
    description = "Enable backward movement"
}

# =====
# LIST VARIABLE
# Multiple ordered directions
# =====

variable "directions" {
    type      = list(string)
    default   = ["left-side", "right-side", "up-side", "down-side"]
    description = "All movement directions"
}

# =====
# MAP VARIABLE
# Key-value movement controls
# =====

variable "car_moves" {
    type = map(string)

    default = {
        l = "left-side"
        r = "right-side"
        u = "up-side"
        d = "down-side"
    }

    description = "Maps control keys to directions"
}

# =====
# OBJECT VARIABLE
# Structured grouped configuration
# =====

variable "car" {
    type = object({
        left = string

```

```

    right = string
    up    = string
    down  = string
  })

  default = {
    left = "left-side"
    right = "right-side"
    up    = "up-side"
    down  = "down-side"
  }

  description = "Complete car movement model"
}

# =====
# TUPLE VARIABLE
# Fixed ordered values
# =====

variable "car_status" {
  type      = tuple([string, string, string, string])
  default    = ["left-side", "right-side", "up-side", "down-side"]
  description = "Fixed directional order"
}

# =====
# SET VARIABLE
# Unique values only
# =====

variable "unique_directions" {
  type      = set(string)
  default    = ["left-side", "right-side", "up-side", "down-side"]
  description = "Unique movement directions"
}

```

Part 2: terraform.tfvars

Purpose:

Overrides default values for real deployment.

```

# STRING VARIABLES
l = "left-side"

```

```

r = "right-side"
u = "up-side"
d = "down-side"

# NUMBER VARIABLES
left_speed = 20
right_speed = 20
up_speed = 60
down_speed = 10

# BOOLEAN VARIABLES
left_enable = true
right_enable = true
up_enable = true
down_enable = true

# LIST VARIABLE
directions = [
    "left-side",
    "right-side",
    "up-side",
    "down-side"
]

# MAP VARIABLE
car_moves = {
    l = "left-side"
    r = "right-side"
    u = "up-side"
    d = "down-side"
}

# OBJECT VARIABLE
car = {
    left = "left-side"
    right = "right-side"
    up = "up-side"
    down = "down-side"
}

# TUPLE VARIABLE
car_status = [
    "left-side",
    "right-side",
    "up-side",
    "down-side"
]

# SET VARIABLE
unique_directions = [
    "left-side",

```

```
"right-side",  
"up-side",  
"down-side"  
]
```

Part 3: Variable Type Comparison Table

Variable Type	Purpose	Car Example	Best Use Case
string	Single value	left-side	Names, locations
number	Numeric value	Speed	VM count, ports
bool	True/False	Engine ON	Feature toggles
list	Ordered values	All directions	Subnets, IP lists
map	Key-value	I = left-side	Tags, configs
object	Structured data	Full car model	Enterprise modules
tuple	Fixed mixed values	Direction order	Strict sequences
set	Unique values	Unique directions	Security rules

Part 4: Key Learning

`variables.tf`

Defines:

- Type
- Default
- Description
- Structure

`terraform.tfvars`

Defines:

- Actual deployment values
- Environment customization
- Production overrides

Real Cloud Example Mapping

Car Concept	Cloud Equivalent
Direction	Resource names
Speed	VM sizing
Enable/Disable	Security controls
List	Multiple subnets
Map	Tags or labels
Object	Full infrastructure module

Beginner Recommendation

Learn in this order:

1. string
2. number
3. bool
4. list
5. map
6. object

This builds strong Terraform architecture skills for:

- Azure
- AWS
- GCP
- DevOps automation
- Enterprise IaC

Final Tip

Use:

- `variables.tf` = blueprint
- `terraform.tfvars` = actual values

Together they create scalable, reusable Terraform deployments.